Volumetric Properties

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Standard Terminology and Volumetric Analysis Used in HMA Technology

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Hot Mix Asphalt
(HMA)
Volumetric Analysis

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Volumetric Analysis

§ All matter has mass and occupies space.

§ Volumetric analysis is a way of evaluating the relationships between mass and volume

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Specific Gravity, G

$$\boxed{Specific \quad Gravity = \left(\frac{Mass}{Volume}\right) / \rho_w}$$

Mass in grams
Volume in cm^3
ρ_w = 1g/cm^3

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Aggregate Specific Gravities

- **Aggregate property relevant to mix volumetrics:**
- **Effective Specific Gravity G_{se}:** is defined as the ratio of the mass in air of a unit volume of a permeable material (**excluding voids permeable to bitumen**) at stated temperature to the mass in air of an equal volume of gas-free distilled water at the stated temperature.

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Bitumen or Binder Content

- In the majority of the design methods, it is expressed by the percentage of the mass of bitumen of by the total mass of the mix.
- Example:
- Assume the mass of aggregate is 1200 g and the mass of bitumen is 63.2g. Determine the bitumen content as a percentage of the total mix.

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Percentage of asphalt by total mass of mix

$$P_b = \frac{63.5}{1200+63.5} * 100 = 5.0\%$$

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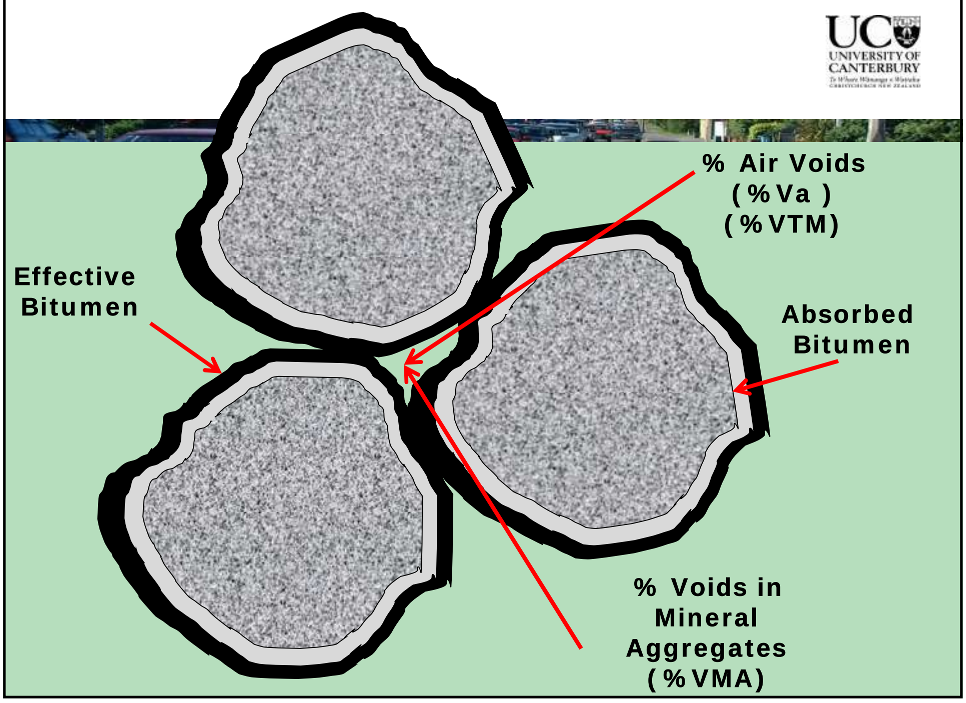

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Voids in the Mineral Aggregate, %VMA

§ The volume of intergranular void space between the aggregate particles of a compacted paving mixture that includes the air voids and the effective bitumen content, expressed as a percent of the total volume of the sample.

Effective Bitumen Content, P_{be}

§ The total bitumen content of a paving mixture minus the portion of bitumen that is lost by absorption into the aggregate particles.

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§ **Air Voids, %V_a or %VTM:** The total volume of the small pockets of air between the coated aggregate particles in a compacted paving mixture, expressed as percent of the bulk volume of the compacted paving mixture.

§ **Voids Filled with Bitumen, %VFB:** The portion of the voids in mineral aggregates (VMA) that is occupied by effective bitumen.

§ **Volume of Voids in the Mineral Aggregate %VMA**: Portion of the compacted mix not occupied by (bulk) aggregates.

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Volumetric Code Names

X_{12}

G = specific gravity
M = mass
W = weight
V = volume
P = percent

b = Bitumen
s = stone (agg)
a = air
m = mix

a = apparent, stone
a = absorbed, binder
b = bulk
e = effective
m = maximum

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Volumetric Abbreviations

- § V_{se} = Effective Volume of Aggregate
- § V_{sb} = Bulk Volume of Aggregate
- § V_b = Total Volume of Bitumen
- § V_{ba} = Volume of Absorbed Bitumen
- § V_{be} = Volume of Effective Bitumen
- § V_{mb} = Total or Bulk Volume of Mix
- § V_{mm} = (Volume that yields maximum specific gravity) i.e,, Voidless Volume of Mix

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Volumetric Abbreviations

- P_b = Percent of all of Bitumen by Total Weight of the Mix
- P_{be} = Percent of Effective Bitumen
- P_{ba} = Percent of Absorbed Bitumen
- P_s = Percent of Aggregate in the Total Mix
- $P_s = (1-P_b)$
- G_{sb} = Bulk Specific Gravity of Aggregate

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Volumetric Abbreviations

- G_{se} = Effective Specific Gravity of Aggregate
- G_{mb} = Bulk Specific Gravity of the Mix
- G_{mm} = Maximum Specific Gravity of the Mix
- %VFB “%VFA”= Voids Filled with Bitumen
- %VA or %VTM= Volume of Air Voids in the Mix
- %VMA = Volume of Voids in the Mineral Aggregate

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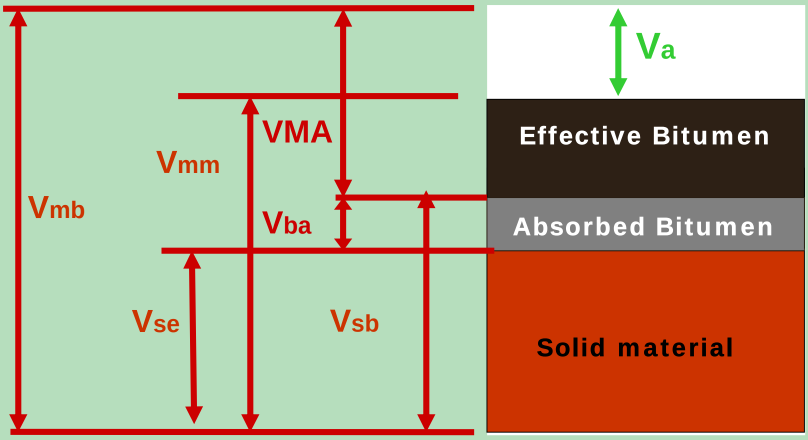
Volumetric Relationships



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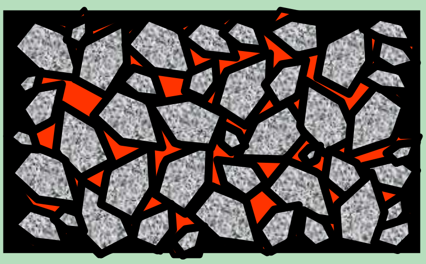
Bulk SG of Compacted HMA

§ AC mixed with agg. and compacted into sample

$$G_{mb} = \frac{\text{Mass of compacted mix}}{\text{Bulk volume of mix}}$$



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Testing

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Testing

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Alternative: Corelok®

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Calculations

§ $G_{mb} = A / (B - C)$

Where:

A = mass of dry compacted specimen

B = mass of SSD compacted specimen

C = mass of submerged compacted spec.

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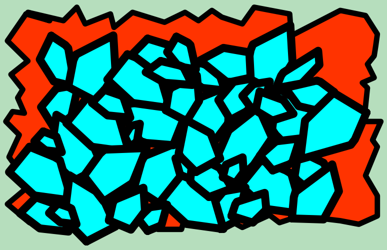
Maximum Theoretical Specific Gravity Gmm

Loose (uncompacted) mixture

G_{mm} = Mass loose mix / Vol. loss mix



$$G_{mm} = \frac{M_m}{V_{mm}}$$

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Maximum Theoretical SP Gr (Gmm)

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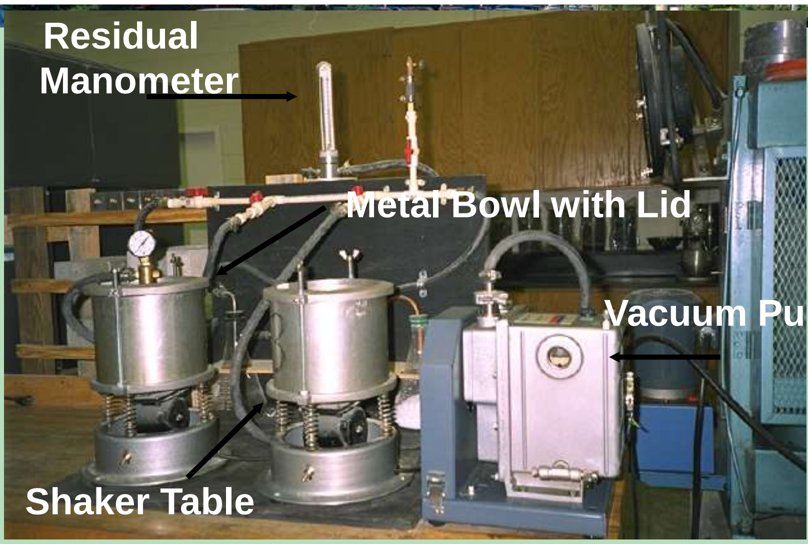
Testing



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Calculations

$$G_{mm} = \frac{(m_3 - m_1)}{(m_3 - m_1) - (m_4 - m_2)}$$

Where:

m_1 = mass of dry Pycnometer

m_2 = mass of dry Pycnometer & water

m_3=mass of dry Pycnometer & loose mix

m_4=mass of dry Pyc & loose mix & water

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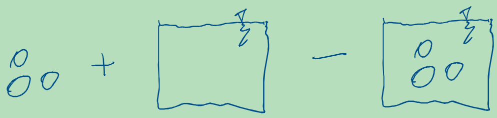

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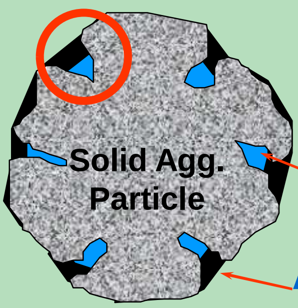
Effective Specific Gravity of Aggregate

Surface Voids



$$G_{se} = \frac{\text{Mass of dry Agg}}{\text{Effective Volume}}$$

Vol. of water-perm. voids not filled with bitumen

Absorbed bitumen

Effective volume = volume of solid aggregate particle + volume of surface voids not filled with bitumen

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% Voids in the Total Mix

%Va = Percent difference between bulk mix volume and Theoretical Max volume

$$\%V_a = \frac{V_{mb} - V_{mm}}{V_{mb}} * 100$$

$$\%Va = \left(1 - \frac{V_{mm}}{V_{mb}} * \frac{M_m}{M_m}\right) * 100$$

$$\%V_a = \left(1 - \frac{G_{mb}}{G_{mm}}\right) * 100$$

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VTM Calculations

§ Determine the percentage of air voids:

G_{mb} = 2.222

G_{mm} = 2.423

%VTM=(1 - 2.222 / 2.423) 100 = 8.3 %

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% Voids in the Mineral Aggregates

%VMA = Percent difference between bulk mix volume minus bulk aggregate volume

$$\% VMA = \frac{V_{mb} - V_{sb}}{V_{mb}} * 100$$

$$\% VMA = \left(1 - \frac{V_{sb}}{V_{mb}} * \frac{M_m}{M_m}\right) * 100$$

$$\% VMA = \left(100 - \frac{V_{sb} * G_{mb} * 100}{M_m}\right)$$

$$M_s = \frac{P_s}{100} * M_m$$

$$M_m = \frac{100 * M_s}{P_s}$$

$$G_{sb} = \frac{M_s}{V_{sb}}$$

$$\% VMA = \left(100 - \frac{P_s * G_{mb}}{G_{sb}}\right)$$

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VMA Calculations

Given that G_{mb} = 2.455, P_s = 95%, and G_{sb} = 2.703

$$\%VMA = \left(100 - \frac{P_s * G_{mb}}{G_{sb}}\right)$$

$$\%VMA = 100 - \frac{(2.455)(95)}{2.703} = 13.7$$

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Effective Aggregate Specific Gravity

$$G_{se}=\frac{M_s}{V_{mm}-V_b}=\frac{(M_m-M_b)/M_m*100}{(V_{mm}-V_b)/M_m*100}=\frac{100-P_b}{\left(\frac{V_{mm}}{M_m}-\frac{V_b}{M_m}*\frac{G_b}{G_b}\right)*100}$$

P_b

$$\boxed{G_{se}=\frac{100-P_b}{\left(\frac{100}{G_{mm}}-\frac{P_b}{G_b}\right)}}$$

Mixture Maximum Specific Gravity

$$G_{mm}=\frac{M_m}{V_{mm}}=\frac{M_m}{\frac{M_s}{G_{se}}+\frac{M_b}{G_b}}=\frac{100}{\frac{P_s}{G_{se}}+\frac{P_b}{G_b}}$$

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Effective Specific Gravity

$$G_{se}=\frac{100-P_b}{\frac{100}{G_{mm}}-\frac{P_b}{G_b}}$$

G_{se} is an aggregate property

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Example Calculations

- P_b (bitumen content)=5.0%
- G_{mm} = 2.535
- G_b = 1.03

$$G_{se} = \frac{100 - 5}{\frac{100}{2.535} - \frac{5}{1.03}} = 2.770$$

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Absolute Bitumen

V_{ba} = Volume of aggregate bulk-volume of aggregate effective

$$V_{ba} = V_{sb} - V_{se} = \frac{W_{agg}}{G_{sb}} - \frac{W_{agg}}{G_{se}} = W_{agg} * (\frac{G_{se} - G_{sb}}{G_{sb} * G_{se}})$$

$$(V_{ba} * G_b) / W_{agg} = (\frac{G_{se} - G_{sb}}{G_{sb} * G_{se}}) * G_b$$

$$\%P_{ba} = (\frac{G_{se} - G_{sb}}{G_{sb} * G_{se}}) * G_b * 100$$

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Percent Binder Absorbed

$$P_{ba} = 100 \left(\frac{G_{se} - G_{sb}}{G_{sb}\ G_{se}} \right) G_b$$

P_{ba} is the percent of absorbed bitumen by mass of aggregate

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Effective Bitumen Content

P_{be} = percent of binder volume not absorbed into aggregate = Percent of binder – percent of binder absorbed

$$P_{be} = P_b - \frac{P_{ba}}{100} * (1 - P_b) = P_b - \frac{P_{ba}}{100} * P_s$$

VMA=(Bulk volume of mix – Bulk volume of aggregates)/Bulk volume of mix *100

$$\%VMA = \frac{V_{mb} - V_{sb}}{V_{mb}} * \frac{W_t}{W_t} * 100 = \left(100 - \frac{\frac{W_t}{V_{mb}}}{\frac{W_t}{V_{sb}}} * 100 \right)$$

$$W_{agg} = P_s * W_t$$

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Percent Air Voids, %VA

$$VA = \frac{\text{Bulk volume} - \text{Voidless volume (vmm)}}{\text{Bulk volume}} * 100$$

$$VA\% = \frac{V_{mb} - V_{mm}}{V_{mb}} * 100 = 100 * \left(1 - \frac{v_{mm}}{v_{mb}} * \frac{W_t}{W_t}\right) = 100 * \left(1 - \frac{G_{mb}}{G_{mm}}\right)$$

- **Percent Air Voids Filled with Asphalt, VFA**

$$VFB\% = \frac{VMA - VA}{VMA} * 100$$

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Effective Asphalt Content

$$P_{be} = P_b - \frac{P_{ba}}{100} (P_s)$$

The effective bitumen content is the total bitumen content minus the percent lost to absorption (based on mass of total mix).

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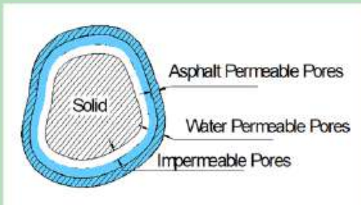
Summary of Vol. Relationships

- Vol. Aggregate Bulk-Vol. Aggregate apparent = Vol. of Water
- Vol. Aggregate Bulk-Vol. Aggregate effective = Vol. binder absorbed
- Vol. binder total-Vol. binder absorbed = Vol. binder effective
- Vol. of mix Bulk-Vol. mix Max = Vol. of air
- Vol. of mix Bulk-Vol. of aggregate Bulk = VMA
- VMA – Vol. Air = Vol. mix filled with binder
- % voids filled = (VMA-Vol. Air)/VMA*100



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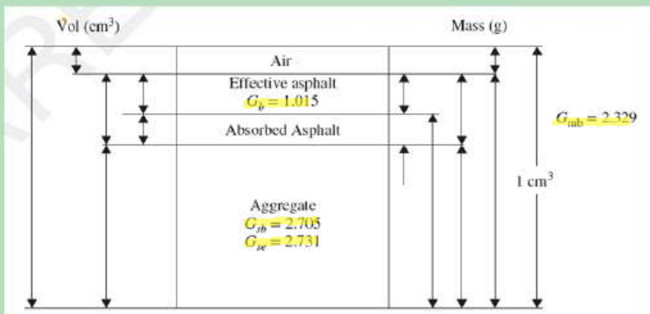
Example

An asphalt mixture has a bulk specific gravity G_{mb} of 2.329. The following phase diagram shows four specific gravities and the asphalt content (P_b = 5%) of a compacted asphalt mixture specimen that have been measured at 25°C. Using only these values, find all the volumetric properties and mass quantities as indicated in the following component diagram.



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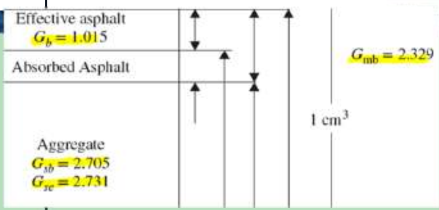
Answer



Based on this information, the values of the component diagram are obtained as follows:

- $G_{mb} = 2.329$, $\gamma_{mb} = G_{mb}\gamma_w = 2.329\,\text{g/cm}^3 = W_{mb}/V_{mb}$
  $W_{mb} = 2.329\,\text{g/cm}^3 \times 1\,\text{cm}^3 = 2.329\,\text{g}$
- $P_b = 5\%$ by mix, $P_b = 0.05 = W_b/W_{mb}$,
  $W_b = 0.05 \times 2.329\,\text{g} = 0.1164\,\text{g}$
- $G_b = 1.015$, $\gamma_b = G_b\gamma_w = 1.015\,\text{g/cm}^3$, $V_b = W_b/\gamma_b = 0.1164\,\text{g}/1.015\,\text{g/cm}^3 = 0.1147\,\text{cm}^3$
- $W_{agg} = W_{mb} - W_b = 2.329 - 0.1164\,\text{g} = 2.2126\,\text{g}$

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Cont.

- $G_{se} = 2.731$, $\gamma_{se} = G_{se}\gamma_w = 2.731\,\text{g/cm}^3 = W_{agg}/V_{se}$
  $V_{se} = W_{agg}/\gamma_{se} = 2.2126\,\text{g}/2.731\,\text{g/cm}^3 = 0.8102\,\text{cm}^3$
- $V_{air} = V_{mb} - V_{se} - V_b = 1 - 0.8102 - 0.11473\,\text{cm}^3 = 0.0751\,\text{cm}^3$ or 7.51%
- $G_{sb} = 2.705$ then, $\gamma_{sb} = G_{sb}\gamma_w = 2.705\,\text{g/cm}^3 = W_{agg}/V_{sb}$,
  $V_{sb} = W_{agg}/\gamma_{sb} = 2.2126\,\text{g}/2.705\,\text{g/cm}^3 = 0.8180\,\text{cm}^3$
- Volume of $VMA = V_{mb} \cancel{+} V_{sb} = (1\text{–}0.8180)\,\text{cm}^3 = 0.1820\,\text{cm}^3$.
  Therefore: $VMA = 18.20\%$
- $V_{ba} = V_{air} + V_b - VMA = (0.0751 + 0.1147 - 0.1820)\,\text{cm}^3 = 0.0078\,\text{cm}^3$
- $V_{be} = V_b - V_{ba} = (0.1147 - 0.007787)\,\text{cm}^3 = 0.1069\,\text{cm}^3$
- $W_{be} = \gamma_{be} \times V_{be} = 1.015\,\text{g/cm}^3 \times 0.1069\,\text{cm}^3 = 0.1085\,\text{g}$
- $W_{ba} = W_b - W_{be} = 0.1164 - 0.1085\,\text{g} = 0.0079\,\text{g}$

complete the initial component

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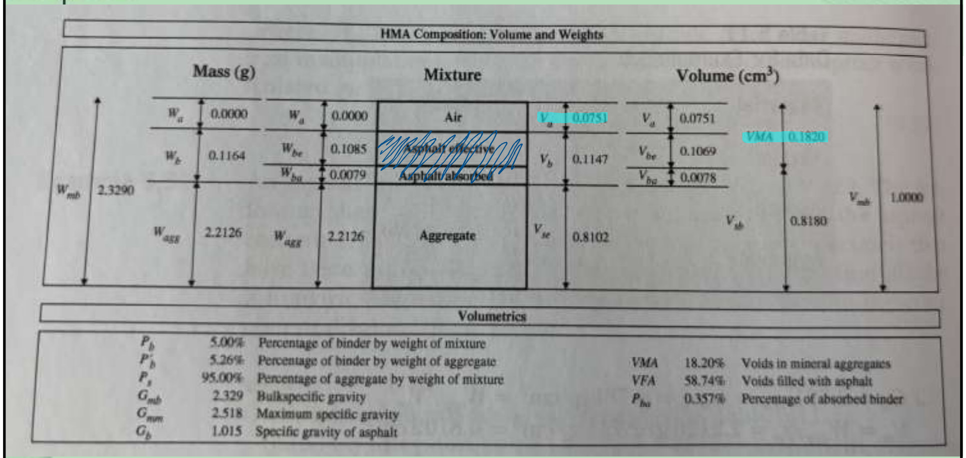
Cont.



P_b	5.00%	Percentage of binder by weight of mixture
P'_b	5.26%	Percentage of binder by weight of aggregate
P_s	95.00%	Percentage of aggregate by weight of mixture
G_{mb}	2.329	Bulkspecific gravity
G_{mm}	2.518	Maximum specific gravity
G_b	1.015	Specific gravity of asphalt
VMA	18.20%	Voids in mineral aggregates
VFA	58.74%	Voids filled with asphalt
P_{ba}	0.357%	Percentage of absorbed binder

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QUESTIONS ?

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